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Roll No.....

EC-503 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Digital Communication
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. With the help of circuit diagram explain ADM (Adaptive Delta Modulation).
2. Explain how PPM and PWM signals are generated from:
 - i) PAM signals
 - ii) DirectlyHow are these detected?
3. A fair die is tossed five times. A toss is called a success if face 1 or 6 appears find
 - i) The probability of 2 successes
 - ii) The mean and standard deviation for the number of successes.
4. Explain the following terms.
 - i) Central limit theorem
 - ii) Auto correlation
 - iii) Probability density function
5. Discuss and characterize Gaussian distribution.
6. With reference to baseband binary transmission. What is an optimum linear receiver? Also draw the block diagram of an optimum linear receiver for binary baseband signalling.
7. Define and explain inter symbol interference. Also describe the following with them use:
 - i) Eye pattern
 - ii) Equalizers

OR

Discuss Bayes theorem.

8. Write short notes on any two of the followings.
 - a) Convolutional codes
 - b) Mutual information
 - c) Nyquist criterion
 - d) Parity codes

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